

University of Nottingham
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Pointers and Input-Output

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Lab Work

- Objectives:
  - Write a Hack assembly program for handling keyboard input and screen output.
  - Write a Hack assembly program for sorting an array.

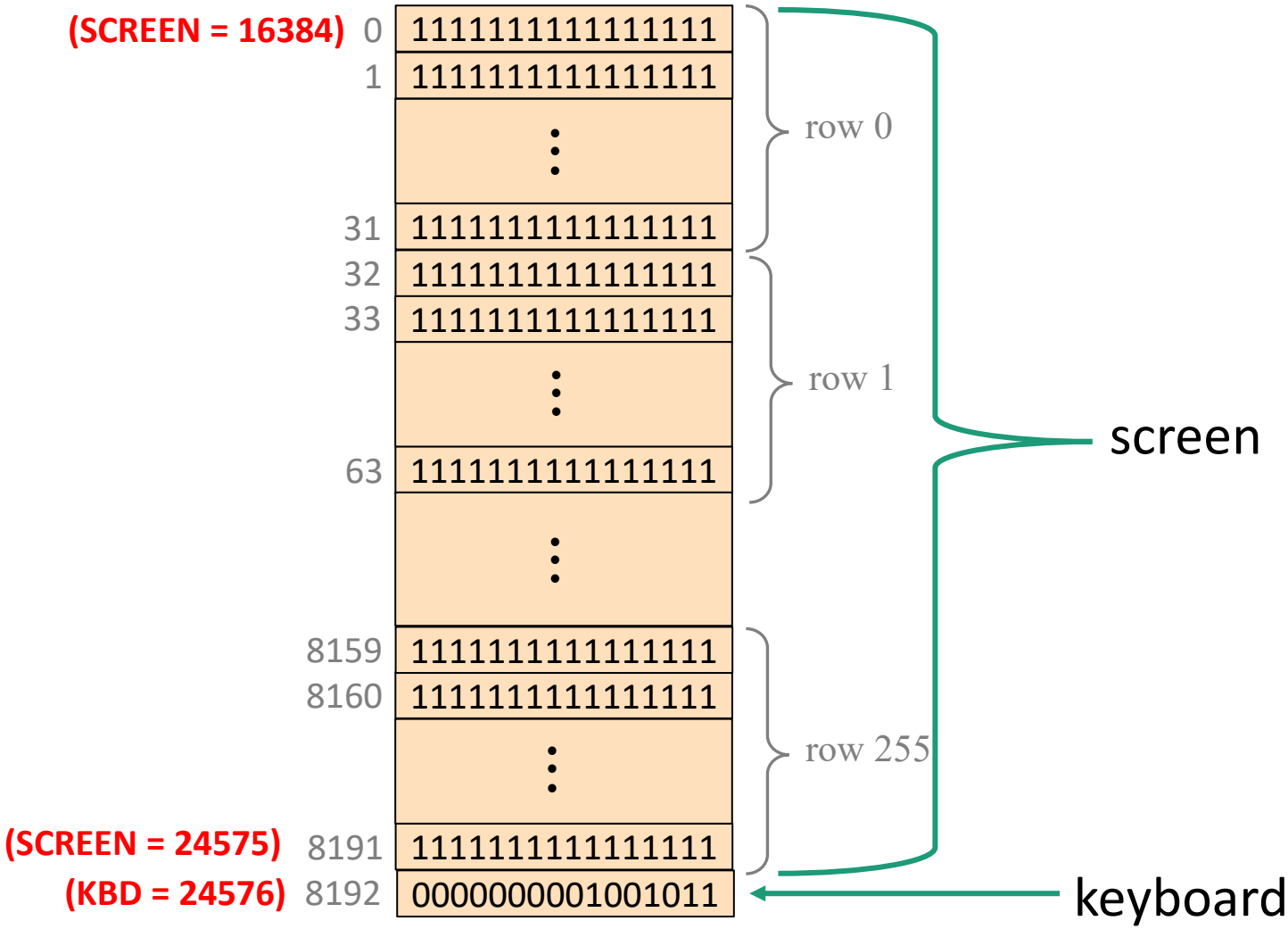
Task 1: fill a rectangle interactively (1)

- `Fill`: a simple interactive program
  - Runs an infinite loop that listens to the keyboard input.
  - When a key is pressed (any key), the program blackens the screen,i.e. writes "black" in every pixel;
  - the screen should remain fully black as long as the key is pressed.
  - When no key is pressed, the program clears the screen, i.e. write "white" in every pixel;
  - the screen should remain fully clear as long as no key is pressed.

Task 1: Fill a Rectangle Interactively (2)

- Implementation strategy
  - Listen to the keyboard
  - To blacken / clear the screen, write code that fills the entire screen memory map with either “white” or “black” pixels
  - Accessing the memory requires working with pointers
  - Hack computer is slow!!!
- Testing
  - Select “no animation”
  - Manual testing (no test scripts).

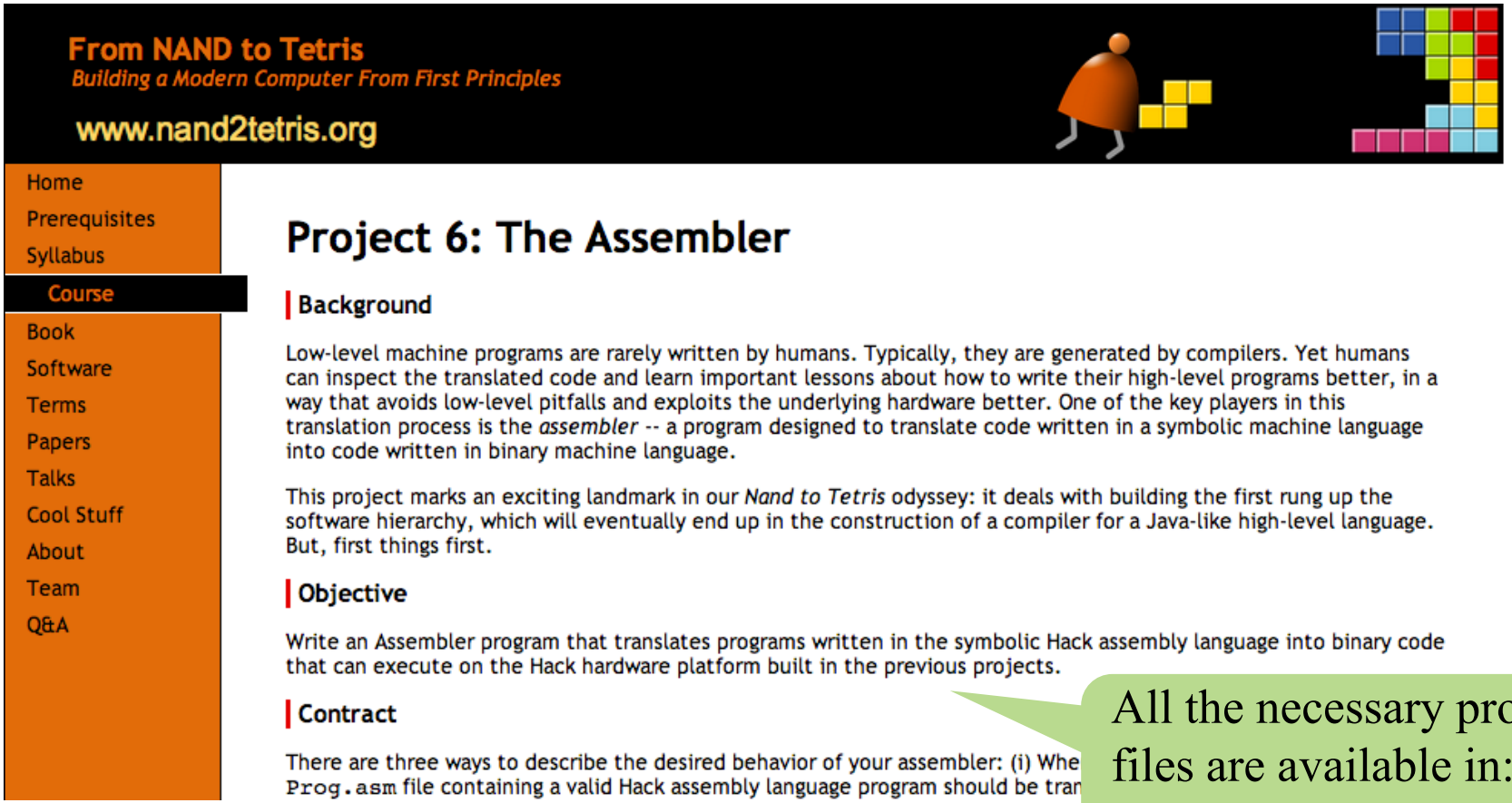
Task 1: Hints


(SCREEN = 16384)
0
1
31
32
33
63
8159
8160
8191
(SCREEN = 24575)
8192
(KBD = 24576)
1111111111111111
row 0
row 1
row 255
screen
keyboard
0000000001001011
scan-code of 'k' = 75

Task 2 – Sorting an Array

- Write a Hack assembly program for sorting an array.
- An unsorted array of integers are given in RAM[11] to RAM[15].
- RAM[0] is used for storing array size.
- RAM[1] to RAM[10] are used for temporary locations for calculations.

Project Resources



All the necessary project files are available in:
nand2tetris / projects / 06

Acknowlegement

- This set of lecture notes are based on the lecture notes provided by Noam Nisam / Shimon Schocken.
- You may find more information on: www.nand2tetris.org.